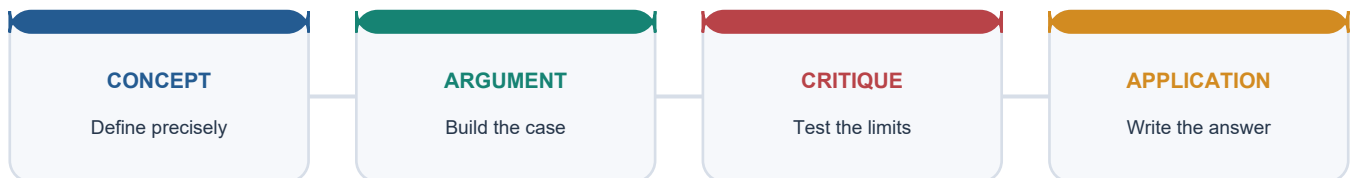


COMPLETE LEARNING SESSION

Climate Economics, Green Finance and Circular Economy - Complete Learning Session

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

Climate Economics, Green Finance and Circular Economy - Complete Learning Session

SOURCE, TIMELINE AND CURRENT-STATUS CONTROL

Current-source cutoff: 10 September 2026.

Mandatory source order used

1. Canonical Economy and Environment Basic/Core Markdown.
2. Canonical Optional Advanced and complete-package Markdown.
3. OCR-searchable *Indian Economy* and *Economic Survey 2025-26* material.
4. Dated official UNFCCC/IPCC, Ministry of Finance/DEA, RBI, SEBI, BEE, MoEFCC, CPCB, MoSPI, NITI Aayog and European Commission sources.
5. Qdrant was not required.

Evidence discipline: A policy announcement, notified rule, project allocation, private capital mobilisation, verified environmental outcome and market price are different facts. Every current threshold, glide path, issuance, draft or operating status below carries its date and perimeter.

OFFICIAL SOURCE REGISTER

- IPCC AR6 glossary and Synthesis Report: <https://www.ipcc.ch/report/ar6/syr/>
- UNFCCC climate-finance introduction and Fund for Responding to Loss and Damage: <https://unfccc.int/topics/introduction-to-climate-finance>
- Energy Conservation (Amendment) Act, 2022 and Carbon Credit Trading Scheme notifications of 28 June and 19 December 2023.
- Bureau of Energy Efficiency, Indian Carbon Market/CCTS institutional and mechanism page: https://beeindia.gov.in/show_content.php?lang=1&level=1&ls_id=189&lid=294
- PIB release ID 2217239 on notified greenhouse-gas emission-intensity targets and obligated entities, January 2026.
- CERC Renewable Energy Certificate Regulations, 2022.
- Department of Economic Affairs, Framework for Sovereign Green Bonds, approved 9 November 2022.
- Department of Economic Affairs, Sovereign Green Bond Allocation Report FY 2023-24.
- Department of Economic Affairs, Draft Framework of India's Climate Finance Taxonomy, press communique dated 7 May 2025.
- RBI, Framework for Acceptance of Green Deposits, 11 April 2023, effective 1 June 2023.
- RBI, sector-specific Climate Finance and Management of Climate Change Risks Directions, 28 November 2025.
- RBI, Draft Disclosure Framework on Climate-related Financial Risks, 28 February 2024, and RBI climate-risk speech of March 2025.
- SEBI, BRSR Core circular of 12 July 2023 and ease-of-doing-business circular dated 28 March 2025.
- SEBI green-debt disclosure and anti-greenwashing circulars dated 3 and 6 February 2023.
- MoSPI, *EnviStats India 2025* and System of Environmental-Economic Accounting material.
- MoEFCC/CPCB waste rules and EPR portals: plastic packaging guidelines dated February 2022; Battery Waste Management Rules, 22 August 2022; E-Waste (Management) Rules, 2 November 2022, effective 1 April 2023; waste-tyre EPR amendment, 21 July 2022; and used-oil EPR amendment, 18 September 2023.
- European Commission, CBAM definitive-regime page, effective 1 January 2026.

ORIGIN AND STATUS TIMELINE

Date	Development	Exam-safe status
1992	UNFCCC adopted	Framework convention; obligations reflect equity and differentiated responsibilities
2015	Paris Agreement adopted	NDC-based mitigation, adaptation and finance architecture
2012 onward	PAT cycles operationalised	Tradable energy-saving certificates linked to energy-intensity targets, not general carbon allowances
2022	Energy Conservation Act amended; CERC REC Regulations, 2022; sovereign green-bond framework approved 9 November	Separate energy-efficiency, renewable-attribute and sovereign use-of-proceeds instruments
2023	CCTS notified 28 June and amended 19 December; RBI green-deposit framework effective 1 June	CCTS contains compliance and offset mechanisms; green deposits are fixed-period earmarked deposits
FY 2022-23	First Indian sovereign green-bond issuance: Rs 16,000 crore	Issuance is borrowing; environmental credibility depends on allocation and impact reporting
FY 2023-24	SGrBs worth Rs 20,000 crore issued	Latest official allocation report used here is for FY 2023-24
28 Feb 2024	RBI issued draft climate-risk disclosure framework	Draft consultation is not a final mandatory disclosure direction
7 May 2025	DEA released draft climate-finance-taxonomy framework	No final taxonomy was identified by the cutoff
FY 2025-26	BRSR Core assessment/assurance glide path covers top 500 listed entities	Ordinary BRSR and BRSR Core assurance are different perimeters
Oct 2025	First GHG-intensity target rules covered 282 entities in four sectors	Legally notified compliance targets; do not infer a mature traded-price history
28 Nov 2025	RBI consolidated green-deposit directions for commercial banks, SFBs and covered deposit-taking NBFCs/HFCs	Immediate directions; the separate February 2024 climate-disclosure framework remained draft at cutoff
13 Jan 2026	Four further industrial categories added, taking the official total to 490 obligated entities	Compliance coverage expanded; market liquidity and price discovery remain developing
1 Jan 2026	EU CBAM definitive regime began	EU import measure, not an Indian carbon tax or part of CCTS
26 Jun 2026	Draft iron-and-steel GHG-intensity targets published for consultation	Draft status retained; no final notification inferred

DATED CURRENT FACT RAIL

Measure or rule	Dated official fact	Qualification
CCTS coverage	PIB, January 2026: 282 entities in the October 2025 first set plus 208 added in January 2026, total 490	Coverage does not prove that an exchange price or mature secondary market already exists
CCTS unit	One Carbon Credit Certificate represents one tonne of CO ₂ -equivalent reduction or removal under the scheme	Certificate integrity still depends on baseline, MRV, additionality and legal eligibility
SGrB FY 2023-24	Rs 20,000 crore issued in four tranches	Debt proceeds are deposited in the Consolidated Fund and tracked for eligible expenditure

Measure or rule	Dated official fact	Qualification
SGrB allocation	FY 2023-24 report: Rs 20,785.60 crore transferred to eligible schemes; Rs 785.60 crore above bond proceeds came from general revenue	Do not add sector allocations and call the total bond issuance
SGrB sector transfers	Rs 16,087.99 crore clean transport; Rs 4,578.47 crore renewable energy; Rs 119.14 crore living resources/land use	Allocation is not independently verified avoided emissions
Climate taxonomy	DEA draft released 7 May 2025; comments invited to 25 June 2025	No final national taxonomy identified by 10 September 2026
Green deposits	RBI directions dated 28 November 2025 consolidated the framework for specified banks and deposit-taking NBFCs/HFCs	Deposit label does not guarantee project success; allocation assurance and impact assessment remain necessary
RBI climate disclosure	Draft issued 28 February 2024; no final notification identified by cutoff	Green-deposit directions do not silently finalise the separate disclosure draft
RBI stress testing	March 2025 RBI speech said scenario-analysis/stress-testing guidance was being developed	Scenario analysis is exploratory, not a point forecast
BRSR Core	SEBI March 2025 circular: top 500 listed entities in FY 2025-26; top 1,000 in FY 2026-27	Value-chain ESG disclosure is voluntary from FY 2025-26; assessment/assurance voluntary from FY 2026-27
CBAM	EU definitive regime began 1 January 2026 for its specified imported goods	Treatment of a carbon price paid abroad is governed by EU rules; recognition of Indian certificates must not be assumed

BASIC / CORE LEARNING SESSION

SESSION 1 - FOUNDATION - Climate change as an economic externality

VISUAL FIRST

PRIVATE BENEFIT / COST
|
UNPRICED EMISSIONS DAMAGE
v
SOCIAL COST > PRIVATE COST
v
OVER-EMISSION WITHOUT POLICY

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Climate change is a negative externality because emitters do not automatically bear the full climate, health and ecosystem damage imposed on others.

Technical definition: Climate change as an economic externality is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

The climate problem begins with a wedge between private cost and social cost.

MUST-WRITE KEYWORDS

- externality
- marginal social damage

- private cost
- social cost
- market failure

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 1:** The climate problem begins with a wedge between private cost and social cost.
- **Named evidence 1:** IPCC assessments connect anthropogenic greenhouse-gas accumulation with widespread physical and economic impacts.
- **Analysis 1:** When damages are unpriced, emissions-intensive production and consumption exceed the socially efficient level.
- **Qualification 1:** Measurement uncertainty does not make the externality zero; it changes the design and precaution needed.

PRELIMS TRAP

An externality is not merely any undesirable side effect; it is an uncompensated effect on others.

MAINS USE

Use the private-cost versus social-cost wedge to introduce carbon-pricing or regulatory answers.

MINI RECAP

Define climate change as an economic externality, name the evidence, trace the mechanism and retain this limit: Measurement uncertainty does not make the externality zero; it changes the design and precaution needed.

SESSION 2 - FOUNDATION - Public bad, commons and free riding

VISUAL FIRST

```

SHARED ATMOSPHERE
|
MANY EMITTERS -> DIFFUSE BENEFITS
|
FREE-RIDER INCENTIVE
v
COOPERATION + VERIFICATION
  
```

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A stable climate has public-good features, while the atmosphere's limited absorptive capacity resembles a global commons vulnerable to overuse.

Technical definition: Public bad, commons and free riding is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Non-excludability creates incentives to free ride, making unilateral action insufficient.

MUST-WRITE KEYWORDS

- public bad
- global commons
- free rider
- collective action
- carbon budget

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 2:** Non-excludability creates incentives to free ride, making unilateral action insufficient.

- **Named evidence 2:** The UNFCCC and Paris Agreement organise collective action through nationally determined commitments, reporting and finance.
- **Analysis 2:** Each country benefits from others' mitigation while bearing the domestic cost of its own action, so cooperation and credibility matter.
- **Qualification 2:** The atmosphere is not a textbook common-pool resource in every respect; the analogy explains rivalry in a finite carbon budget.

PRELIMS TRAP

Do not call climate change a purely local pollution problem.

MAINS USE

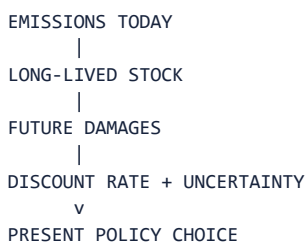
Link global collective action to equity, finance and differentiated capabilities.

MINI RECAP

Define public bad, commons and free riding, name the evidence, trace the mechanism and retain this limit: The atmosphere is not a textbook common-pool resource in every respect; the analogy explains rivalry in a finite carbon budget.

SESSION 3 - FOUNDATION - Intergenerational equity and uncertainty

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Intergenerational climate economics studies how present choices redistribute costs, risks and opportunities across future generations.

Technical definition: Intergenerational equity and uncertainty is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Long-lived emissions make today's discount rate an ethical as well as technical choice.

MUST-WRITE KEYWORDS

- intergenerational equity
- discounting
- uncertainty
- irreversibility
- precaution

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 3:** Long-lived emissions make today's discount rate an ethical as well as technical choice.
- **Named evidence 3:** IPCC scenarios show that cumulative emissions shape long-term warming and associated risks.
- **Analysis 3:** A higher discount rate places less present weight on distant harm, while uncertainty and irreversible thresholds can support precaution.

- **Qualification 3:** No single discount rate or damage path is universally correct; assumptions must be disclosed.

PRELIMS TRAP

Discounting does not mean future lives are valueless.

MAINS USE

Use uncertainty, irreversibility and option value to qualify cost-benefit analysis.

MINI RECAP

Define intergenerational equity and uncertainty, name the evidence, trace the mechanism and retain this limit: No single discount rate or damage path is universally correct; assumptions must be disclosed.

SESSION 4 - FOUNDATION - Mitigation, adaptation and loss and damage

VISUAL FIRST

MITIGATION -> LOWER FUTURE HAZARD
ADAPTATION -> LOWER EXPOSURE / VULNERABILITY
RESIDUAL IMPACT -> LOSS AND DAMAGE
ALL THREE -> DIFFERENT FINANCE NEEDS

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Mitigation reduces emissions or enhances sinks; adaptation reduces exposure or vulnerability; loss and damage concerns adverse effects not avoided through mitigation and adaptation.

Technical definition: Mitigation, adaptation and loss and damage is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

The three responses address different points in the climate-risk chain and cannot substitute perfectly for one another.

MUST-WRITE KEYWORDS

- mitigation
- adaptation
- resilience
- loss and damage
- residual risk

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 4:** The three responses address different points in the climate-risk chain and cannot substitute perfectly for one another.
- **Named evidence 4:** IPCC and UNFCCC materials distinguish emission reduction, adjustment to impacts and residual economic or non-economic loss.
- **Analysis 4:** Early mitigation limits future hazard, adaptation reduces realised harm, and loss-and-damage arrangements respond to residual impacts.
- **Qualification 4:** Some measures provide both mitigation and adaptation benefits, but their finance and outcome metrics remain distinct.

PRELIMS TRAP

Do not label disaster relief as mitigation or assume adaptation eliminates residual loss.

MAINS USE

Structure answers as hazard reduction, vulnerability reduction and residual-risk response.

MINI RECAP

Define mitigation, adaptation and loss and damage, name the evidence, trace the mechanism and retain this limit: Some measures provide both mitigation and adaptation benefits, but their finance and outcome metrics remain distinct.

SESSION 5 - FOUNDATION - Social Cost of Carbon

VISUAL FIRST

ONE EXTRA tCO₂e
|
FUTURE DAMAGE STREAM
|
DISCOUNT + DAMAGE MODEL
v
SOCIAL COST OF CARBON
(APPRAISAL VALUE)

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: The Social Cost of Carbon is the discounted monetary estimate of incremental damages caused by one additional tonne of carbon-dioxide-equivalent emissions.

Technical definition: Social Cost of Carbon is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

SCC is an appraisal shadow value, not automatically a tax rate or market price.

MUST-WRITE KEYWORDS

- SCC
- marginal damage
- present value
- discount rate
- damage function

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 5:** SCC is an appraisal shadow value, not automatically a tax rate or market price.
- **Named evidence 5:** The 2020 UPSC objective route tests SCC as monetary valuation of emissions damage.
- **Analysis 5:** It translates future climate and health damages into a present marginal value for comparing policies and projects.
- **Qualification 5:** Results vary with damage functions, climate sensitivity, socioeconomic pathways and discount rates; no universal number is used here.

PRELIMS TRAP

SCC, carbon tax and traded certificate price are three distinct values.

MAINS USE

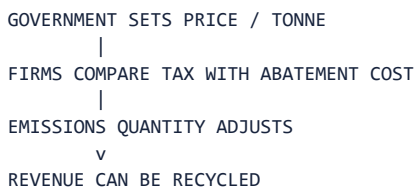
Use SCC to discuss appraisal, then disclose uncertainty and ethical assumptions.

MINI RECAP

Define social cost of carbon, name the evidence, trace the mechanism and retain this limit: Results vary with damage functions, climate sensitivity, socioeconomic pathways and discount rates; no universal number is used here.

SESSION 6 - FOUNDATION - Carbon tax design

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A carbon tax imposes a government-set levy per unit of covered emissions or carbon content.

Technical definition: Carbon tax design is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

A tax fixes the marginal price signal while leaving the emissions quantity responsive to behaviour and technology.

MUST-WRITE KEYWORDS

- carbon tax
- price certainty
- tax base
- pass-through
- revenue recycling

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 6:** A tax fixes the marginal price signal while leaving the emissions quantity responsive to behaviour and technology.
- **Named evidence 6:** Tax design requires a taxable base, rate, point of collection, monitoring and treatment of imports, exemptions and revenue.
- **Analysis 6:** A broad predictable price can reduce abatement-cost differences, while revenue recycling can protect households or fund transition.
- **Qualification 6:** Demand uncertainty means the resulting emissions quantity is not predetermined.

PRELIMS TRAP

A fuel excise is not automatically a comprehensive carbon tax unless its base and purpose match.

MAINS USE

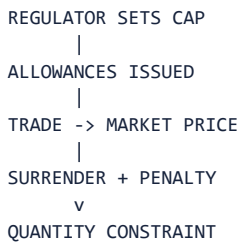
Evaluate efficiency together with regressivity, competitiveness, administration and revenue use.

MINI RECAP

Define carbon tax design, name the evidence, trace the mechanism and retain this limit: Demand uncertainty means the resulting emissions quantity is not predetermined.

SESSION 7 - FOUNDATION - Cap-and-trade

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Cap-and-trade fixes an aggregate emissions cap and creates tradable allowances whose price emerges from supply and demand.

Technical definition: Cap-and-trade is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Trading equalises marginal abatement costs only when the cap, monitoring and market design are credible.

MUST-WRITE KEYWORDS

- cap
- allowance
- auction
- banking
- marginal abatement cost
- leakage

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 7:** Trading equalises marginal abatement costs only when the cap, monitoring and market design are credible.
- **Named evidence 7:** Allowance systems specify covered sources, allocation or auction, banking, compliance and penalties.
- **Analysis 7:** Low-cost reducers sell allowances while high-cost reducers buy, achieving the cap at lower aggregate cost.
- **Qualification 7:** Price volatility, overallocation, market power and leakage can weaken performance.

PRELIMS TRAP

A tradable allowance under a cap is not the same as a project offset.

MAINS USE

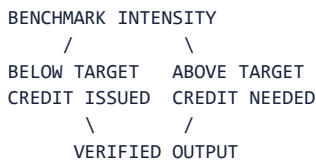
Compare quantity certainty against price uncertainty and add a price-stability mechanism if relevant.

MINI RECAP

Define cap-and-trade, name the evidence, trace the mechanism and retain this limit: Price volatility, overallocation, market power and leakage can weaken performance.

SESSION 8 - FOUNDATION - Baseline-and-credit systems

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A baseline-and-credit system awards credits when verified performance beats a benchmark and may require covered entities above the benchmark to acquire credits.

Technical definition: Baseline-and-credit systems is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

An intensity benchmark can preserve output flexibility but may allow absolute emissions to rise.

MUST-WRITE KEYWORDS

- **baseline**
- **credit**
- **intensity target**
- **equivalent product**
- **absolute emissions**

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 8:** An intensity benchmark can preserve output flexibility but may allow absolute emissions to rise.
- **Named evidence 8:** India's CCTS compliance mechanism sets greenhouse-gas emission intensity targets per unit of equivalent product.
- **Analysis 8:** Credits reward performance below the target and purchases cover the shortfall, creating an output-linked incentive.
- **Qualification 8:** Baseline choice and output measurement strongly affect credit supply and environmental integrity.

PRELIMS TRAP

Baseline-and-credit is not automatically an absolute cap-and-trade system.

MAINS USE

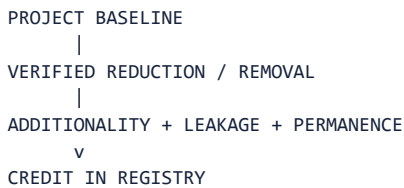
State whether the obligation concerns absolute tonnes or intensity per unit of output.

MINI RECAP

Define baseline-and-credit systems, name the evidence, trace the mechanism and retain this limit: Baseline choice and output measurement strongly affect credit supply and environmental integrity.

SESSION 9 - FOUNDATION - Offsets and credit integrity

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: An offset is a verified project-based reduction, avoidance or removal relative to a counterfactual baseline, used against emissions elsewhere.

Technical definition: Offsets and credit integrity is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Offset quality rests on additionality, conservative baselines, leakage control, permanence and no double counting.

MUST-WRITE KEYWORDS

- offset
- counterfactual
- additionality
- leakage
- permanence
- double counting

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 9:** Offset quality rests on additionality, conservative baselines, leakage control, permanence and no double counting.
- **Named evidence 9:** The December 2023 CCTS amendment created an offset mechanism for non-obligated entities using approved methodologies and verification.
- **Analysis 9:** A credit is credible only if the claimed reduction would not otherwise occur and remains attributable over the stated period.
- **Qualification 9:** Avoided emissions and removals have different reversal and measurement risks.

PRELIMS TRAP

Registration or a certificate does not by itself prove additionality.

MAINS USE

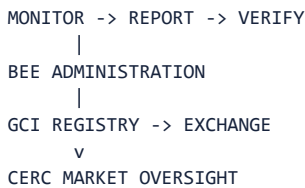
Audit baseline, additionality, leakage, permanence, registry and corresponding adjustment where applicable.

MINI RECAP

Define offsets and credit integrity, name the evidence, trace the mechanism and retain this limit: Avoided emissions and removals have different reversal and measurement risks.

SESSION 10 - FOUNDATION - MRV and market integrity

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Monitoring, reporting and verification is the measurement and assurance chain that supports compliance and credit issuance.

Technical definition: MRV and market integrity is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Carbon markets are only as credible as their data, accreditation, registry and enforcement.

MUST-WRITE KEYWORDS

- MRV
- ACVA
- registry
- CERC
- surrender
- environmental integrity

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 10:** Carbon markets are only as credible as their data, accreditation, registry and enforcement.
- **Named evidence 10:** BEE administers CCTS, accredited carbon verification agencies validate evidence, Grid Controller of India operates the registry and CERC regulates trading.
- **Analysis 10:** Separated institutional roles reduce conflicts and create traceability from reported emissions to certificate surrender.
- **Qualification 10:** Verification lowers information risk but cannot remove uncertainty in every baseline or emission factor.

PRELIMS TRAP

A registry entry is not equivalent to independent proof of real-world climate benefit.

MAINS USE

Map administrator, verifier, registry, exchange regulator and obligated entity.

MINI RECAP

Define mrv and market integrity, name the evidence, trace the mechanism and retain this limit: Verification lowers information risk but cannot remove uncertainty in every baseline or emission factor.

SESSION 11 - FOUNDATION - PAT energy-efficiency market

VISUAL FIRST

DESIGNATED CONSUMER
|
SPECIFIC ENERGY TARGET
|
OVERACHIEVE -> ESCERT
UNDERACHIEVE -> BUY / COMPLY

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Perform, Achieve and Trade assigns specific energy-consumption targets to designated consumers and allows trading of Energy Saving Certificates.

Technical definition: PAT energy-efficiency market is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

PAT prices verified energy-efficiency overachievement, not economy-wide greenhouse-gas emissions.

MUST-WRITE KEYWORDS

- PAT
- designated consumer
- SEC
- ESCert
- energy intensity

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 11:** PAT prices verified energy-efficiency overachievement, not economy-wide greenhouse-gas emissions.
- **Named evidence 11:** PAT operates under the Energy Conservation Act and BEE reports cumulative savings through its cycles.
- **Analysis 11:** It built measurement and certificate-trading capacity later used in India's carbon-market architecture.
- **Qualification 11:** Energy-intensity improvement can coexist with rising absolute energy use or emissions when output expands.

PRELIMS TRAP

One ESCert is not interchangeable by definition with one REC or one CCTS carbon credit.

MAINS USE

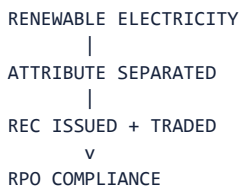
Use PAT as institutional lineage, then identify its energy-intensity boundary.

MINI RECAP

Define pat energy-efficiency market, name the evidence, trace the mechanism and retain this limit: Energy-intensity improvement can coexist with rising absolute energy use or emissions when output expands.

SESSION 12 - FOUNDATION - Renewable Energy Certificates

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A Renewable Energy Certificate represents the environmental attribute of eligible renewable electricity under CERC's regulatory framework.

Technical definition: Renewable Energy Certificates is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

REC supports renewable-purchase compliance and is not a general greenhouse-gas offset.

MUST-WRITE KEYWORDS

- REC
- renewable attribute
- RPO
- CERC
- electricity
- certificate

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 12:** REC supports renewable-purchase compliance and is not a general greenhouse-gas offset.
- **Named evidence 12:** CERC Renewable Energy Certificate Regulations, 2022 govern issuance and dealing in RECs.
- **Analysis 12:** Separating electricity from its renewable attribute enables obligated buyers to meet renewable-purchase requirements.
- **Qualification 12:** A REC does not automatically quantify economy-wide avoided emissions because grid factors and additionality vary.

PRELIMS TRAP

REC, ESCert and Carbon Credit Certificate are legally and functionally distinct.

MAINS USE

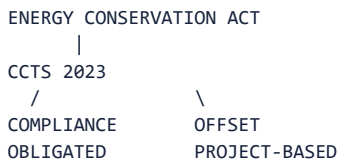
Use a three-column comparison: renewable attribute, energy saving, carbon reduction/removal.

MINI RECAP

Define renewable energy certificates, name the evidence, trace the mechanism and retain this limit: A REC does not automatically quantify economy-wide avoided emissions because grid factors and additionality vary.

SESSION 13 - FOUNDATION - CCTS legal architecture

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: The Carbon Credit Trading Scheme is India's statutory carbon-market framework under the Energy Conservation Act, with compliance and offset mechanisms.

Technical definition: CCTS legal architecture is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

CCTS is a phased baseline-and-credit architecture, not an economy-wide absolute emissions cap.

MUST-WRITE KEYWORDS

- CCTS
- compliance mechanism
- offset mechanism
- BEE
- GCI
- CERC

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 13:** CCTS is a phased baseline-and-credit architecture, not an economy-wide absolute emissions cap.
- **Named evidence 13:** The scheme was notified on 28 June 2023 and amended on 19 December 2023; BEE is administrator, GCI registry operator and CERC trading regulator.
- **Analysis 13:** Obligated entities face product-linked GHG-intensity targets while non-obligated projects may use approved offset methodologies.
- **Qualification 13:** Legal architecture, target notification, registry readiness, credit issuance and liquid trading are separate implementation stages.

PRELIMS TRAP

Do not report a carbon price or trading volume without an official dated market record.

MAINS USE

Begin with legal authority and roles before evaluating incentives.

MINI RECAP

Define ccts legal architecture, name the evidence, trace the mechanism and retain this limit: Legal architecture, target notification, registry readiness, credit issuance and liquid trading are separate implementation stages.

SESSION 14 - FOUNDATION - CCTS current coverage

VISUAL FIRST

OCT 2025 -> 282 ENTITIES / 4 SECTORS
JAN 2026 -> +208 / 4 CATEGORIES
TOTAL -> 490 NOTIFIED
JUN 2026 STEEL -> DRAFT

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Current CCTS coverage is defined by notified entity-level greenhouse-gas intensity targets, not by a broad list of sectors proposed for future inclusion.

Technical definition: CCTS current coverage is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

India had notified 490 obligated entities across eight industrial categories by January 2026.

MUST-WRITE KEYWORDS

- obligated entity
- GHG intensity
- notified target
- compliance year
- draft status

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 14:** India had notified 490 obligated entities across eight industrial categories by January 2026.
- **Named evidence 14:** PIB reported 282 entities in aluminium, cement, chlor-alkali and pulp-and-paper in October 2025, followed by 208 in petroleum refineries, petrochemicals, textiles and secondary aluminium on 13 January 2026.
- **Analysis 14:** The notifications convert framework design into enforceable intensity obligations for specified plants.
- **Qualification 14:** Iron-and-steel targets published on 26 June 2026 remained a draft in the evidence checked for this package.

PRELIMS TRAP

A proposed sector or draft target is not a final obligation.

MAINS USE

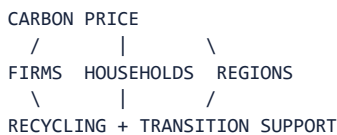
Date the notification, sector set, entity count and compliance year; avoid claiming mature trading.

MINI RECAP

Define ccts current coverage, name the evidence, trace the mechanism and retain this limit: Iron-and-steel targets published on 26 June 2026 remained a draft in the evidence checked for this package.

SESSION 15 - CORE - Carbon-price distribution and competitiveness

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Carbon pricing changes relative prices across fuels, products, households, firms and regions.

Technical definition: Carbon-price distribution and competitiveness is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Efficiency requires complementary distribution and competitiveness design.

MUST-WRITE KEYWORDS

- incidence
- pass-through
- regressivity
- leakage
- competitiveness
- recycling

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 15:** Efficiency requires complementary distribution and competitiveness design.
- **Named evidence 15:** Energy expenditure takes a larger budget share for many vulnerable households, while emissions-intensive trade-exposed firms face leakage risk.
- **Analysis 15:** Targeted transfers, revenue recycling, transition support and border-compatible measurement can preserve incentives while limiting hardship.
- **Qualification 15:** Broad exemptions weaken the price signal and often reward incumbent emissions.

PRELIMS TRAP

Regressivity of energy prices does not imply climate damages are distributionally neutral.

MAINS USE

Pair polluter-pays incentives with transparent use of revenues and time-bound adjustment support.

MINI RECAP

Define carbon-price distribution and competitiveness, name the evidence, trace the mechanism and retain this limit: Broad exemptions weaken the price signal and often reward incumbent emissions.

SESSION 16 - CORE - Climate-policy instrument mix

VISUAL FIRST

EXTERNALITY -> CARBON PRICE
NETWORK GAP -> PUBLIC INVESTMENT
INNOVATION GAP -> R&D SUPPORT
EQUITY GAP -> SOCIAL PROTECTION

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A climate-policy mix combines prices, standards, innovation support, public investment, information and social protection.

Technical definition: Climate-policy instrument mix is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

No single instrument corrects every emissions, technology, network, information and distribution failure.

MUST-WRITE KEYWORDS

- policy mix
- standards
- innovation spillover
- public investment
- social protection

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 16:** No single instrument corrects every emissions, technology, network, information and distribution failure.
- **Named evidence 16:** Carbon pricing addresses operating incentives; standards handle monitoring or behavioural limits; public investment supplies grids and resilience; innovation support addresses knowledge spillovers.
- **Analysis 16:** Complementarity can accelerate transition when mandates, finance and infrastructure reinforce one another.
- **Qualification 16:** Overlapping instruments can also raise cost or create contradictory signals unless responsibilities and timelines align.

PRELIMS TRAP

More instruments do not automatically mean stronger policy.

MAINS USE

Match each failure to an instrument and evaluate interaction, not labels.

MINI RECAP

Define climate-policy instrument mix, name the evidence, trace the mechanism and retain this limit: Overlapping instruments can also raise cost or create contradictory signals unless responsibilities and timelines align.

SESSION 17 - CORE - Climate-finance definition and perimeter

VISUAL FIRST

PUBLIC / PRIVATE / ALTERNATIVE SOURCES
|
MITIGATION OR ADAPTATION PURPOSE
|
ELIGIBILITY + TRACKING + IMPACT
v
CREDIBLE CLIMATE FINANCE

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Climate finance is local, national or transnational finance from public, private or alternative sources supporting mitigation and adaptation.

Technical definition: Climate-finance definition and perimeter is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Purpose and environmental integrity, not an ESG label alone, determine whether finance is climate finance.

MUST-WRITE KEYWORDS

- climate finance
- mitigation finance
- adaptation finance
- eligible activity
- integrity

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 17:** Purpose and environmental integrity, not an ESG label alone, determine whether finance is climate finance.
- **Named evidence 17:** UNFCCC uses a broad source-and-purpose definition linked to mitigation and adaptation actions.
- **Analysis 17:** The same instrument can finance genuinely aligned, transitional or mislabelled assets depending on eligibility, use of proceeds and verification.
- **Qualification 17:** Development finance may produce climate co-benefits without being wholly counted as climate finance.

PRELIMS TRAP

Every ESG fund, ordinary infrastructure loan or renewable-company share is not automatically green finance.

MAINS USE

Define source, instrument, recipient, eligible activity and climate objective.

MINI RECAP

Define climate-finance definition and perimeter, name the evidence, trace the mechanism and retain this limit: Development finance may produce climate co-benefits without being wholly counted as climate finance.

SESSION 18 - CORE - Provided, mobilised and committed finance

VISUAL FIRST

PLEDGE -> COMMITMENT -> APPROVAL
-> DISBURSEMENT -> EXPENDITURE
PUBLIC INPUT -> MOBILISED PRIVATE FINANCE
KEEP FLOWS DISTINCT

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Provided finance is supplied directly by a public actor, while mobilised private finance is attributed to public intervention under a stated methodology.

Technical definition: Provided, mobilised and committed finance is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Commitment, approval, mobilisation, disbursement and expenditure must be reported separately.

MUST-WRITE KEYWORDS

- provided
- mobilised
- committed
- disbursed
- spent
- attribution

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 18:** Commitment, approval, mobilisation, disbursement and expenditure must be reported separately.
- **Named evidence 18:** UNFCCC finance reporting distinguishes sources and channels, while official project reports separately track allocation and impact.
- **Analysis 18:** Separating stages prevents announced envelopes or leveraged private amounts from being treated as cash received by projects.
- **Qualification 18:** Mobilisation estimates depend on attribution rules and may not represent wholly additional private finance.

PRELIMS TRAP

Do not add public finance and the full project value if that double counts mobilisation.

MAINS USE

Use a finance pipeline from pledge to verified outcome.

MINI RECAP

Define provided, mobilised and committed finance, name the evidence, trace the mechanism and retain this limit: Mobilisation estimates depend on attribution rules and may not represent wholly additional private finance.

SESSION 19 - CORE - Additionality and concessionality

VISUAL FIRST

MARKET BARRIER
|
CONCESSIONAL PUBLIC SUPPORT
|
PRIVATE CAPITAL MOBILISED?
|
COUNTERFACTUAL + OUTCOME TEST

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Financial additionality asks whether support causes investment beyond the counterfactual; concessionality means terms more favourable than market finance.

Technical definition: Additionality and concessionality is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Cheap finance is useful only when it addresses a real barrier rather than subsidising activity that would occur anyway.

MUST-WRITE KEYWORDS

- **additionality**
- **concessionality**
- **counterfactual**
- **subsidy element**
- **crowding in**

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 19:** Cheap finance is useful only when it addresses a real barrier rather than subsidising activity that would occur anyway.
- **Named evidence 19:** Public climate funds use grants, concessional debt, guarantees and technical assistance to improve project viability.
- **Analysis 19:** Risk absorption can crowd in private capital for first-of-kind, adaptation or low-income projects.
- **Qualification 19:** Excess subsidy can create rents, moral hazard or technology lock-in.

PRELIMS TRAP

Private co-financing alone does not prove additionality.

MAINS USE

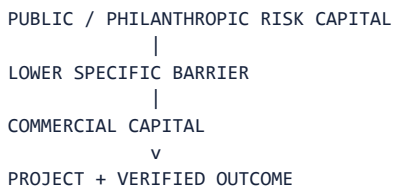
State the barrier, counterfactual, subsidy element, leverage method and exit condition.

MINI RECAP

Define additionality and concessionality, name the evidence, trace the mechanism and retain this limit: Excess subsidy can create rents, moral hazard or technology lock-in.

SESSION 20 - CORE - Blended finance

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Blended finance strategically uses public or philanthropic capital to improve the risk-return profile of projects and mobilise commercial finance.

Technical definition: Blended finance is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Blending should allocate risk to the party best able to manage it, not merely socialise losses.

MUST-WRITE KEYWORDS

- **blended finance**
- **first loss**
- **guarantee**
- **leverage**
- **contingent liability**

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 20:** Blending should allocate risk to the party best able to manage it, not merely socialise losses.
- **Named evidence 20:** First-loss capital, guarantees, viability support and technical assistance are common structures.
- **Analysis 20:** Risk mitigation can make uncertain technologies or underserved markets financeable and create a performance record.
- **Qualification 20:** Opaque guarantees create contingent liabilities and may reward weak projects.

PRELIMS TRAP

A public guarantee does not remove the underlying project or climate risk.

MAINS USE

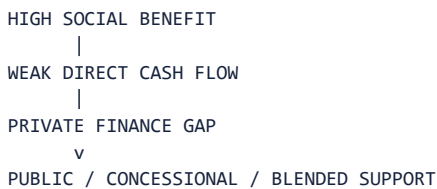
Evaluate leverage, additionality, fiscal risk, pricing, governance and developmental outcome.

MINI RECAP

Define blended finance, name the evidence, trace the mechanism and retain this limit: Opaque guarantees create contingent liabilities and may reward weak projects.

SESSION 21 - CORE - Why adaptation is hard to finance

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Adaptation finance funds investments that reduce exposure or vulnerability to climate impacts.

Technical definition: Why adaptation is hard to finance is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Many adaptation benefits are public, local, uncertain and difficult to monetise, weakening direct revenue streams.

MUST-WRITE KEYWORDS

- adaptation gap
- public good
- avoided loss
- bankability
- resilience dividend

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 21:** Many adaptation benefits are public, local, uncertain and difficult to monetise, weakening direct revenue streams.
- **Named evidence 21:** Flood protection, heat-health systems and climate information create avoided losses shared across communities.
- **Analysis 21:** Grants, public budgets, concessional loans, insurance support and resilience-linked infrastructure finance can fill the bankability gap.
- **Qualification 21:** Avoided-loss estimates depend on hazards, baselines and counterfactuals; debt can burden vulnerable borrowers.

PRELIMS TRAP

Low private investment does not imply low social return.

MAINS USE

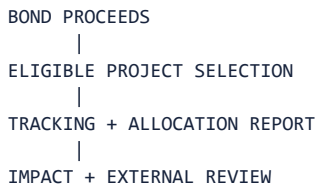
Explain the revenue gap, then propose public/concessional finance and measurable resilience outcomes.

MINI RECAP

Define why adaptation is hard to finance, name the evidence, trace the mechanism and retain this limit: Avoided-loss estimates depend on hazards, baselines and counterfactuals; debt can burden vulnerable borrowers.

SESSION 22 - CORE - Green bonds and use of proceeds

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A green bond is debt whose proceeds are earmarked for eligible environmental projects and tracked under a disclosed framework.

Technical definition: Green bonds and use of proceeds is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

The bond finances an issuer; the green label governs proceeds, reporting and review rather than eliminating credit risk.

MUST-WRITE KEYWORDS

- green bond
- use of proceeds
- ring-fencing
- allocation report
- external review

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 22:** The bond finances an issuer; the green label governs proceeds, reporting and review rather than eliminating credit risk.
- **Named evidence 22:** SEBI's green-debt framework requires environmental objectives, project selection, tracking, disclosures and safeguards against greenwashing.
- **Analysis 22:** Ring-fencing and reporting improve traceability between borrowing and eligible expenditure.
- **Qualification 22:** Allocation evidence is not the same as verified environmental additionality or impact.

PRELIMS TRAP

Green bonds are not automatically safer or cheaper than comparable conventional debt.

MAINS USE

Separate issuer credit risk, use-of-proceeds integrity and measured impact.

MINI RECAP

Define green bonds and use of proceeds, name the evidence, trace the mechanism and retain this limit: Allocation evidence is not the same as verified environmental additionality or impact.

SESSION 23 - CORE - India's sovereign green bonds

VISUAL FIRST

SOVEREIGN BORROWING
|
CONSOLIDATED FUND OF INDIA
|
TRACKED ELIGIBLE EXPENDITURE
v
ALLOCATION + IMPACT REPORT

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A sovereign green bond is government debt issued under a framework that earmarks an equivalent amount for eligible public green expenditure.

Technical definition: India's sovereign green bonds is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Sovereign labelling changes expenditure tracking, not the government's repayment obligation.

MUST-WRITE KEYWORDS

- SGrB
- GFWC
- PDMC
- Consolidated Fund
- eligible expenditure

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 23:** Sovereign labelling changes expenditure tracking, not the government's repayment obligation.
- **Named evidence 23:** India's framework was approved on 9 November 2022; first issuance was Rs 16,000 crore in FY 2022-23 and FY 2023-24 issuance was Rs 20,000 crore.
- **Analysis 23:** The Green Finance Working Committee selects eligible expenditure, the Public Debt Management Cell tracks proceeds, and annual allocation reporting supports accountability.
- **Qualification 23:** Proceeds enter the Consolidated Fund of India; earmarking does not create a separate sovereign credit pool.

PRELIMS TRAP

Do not call total eligible expenditure the bond issuance amount.

MAINS USE

Trace framework, issuance, Consolidated Fund, allocation, impact report and external review.

MINI RECAP

Define India's sovereign green bonds, name the evidence, trace the mechanism and retain this limit: Proceeds enter the Consolidated Fund of India; earmarking does not create a separate sovereign credit pool.

SESSION 24 - CORE - Sovereign allocation evidence

VISUAL FIRST

ISSUED: Rs 20,000 CRORE
ELIGIBLE TRANSFER: Rs 20,785.60 CRORE
DIFFERENCE: Rs 785.60 CRORE
SOURCE: GENERAL REVENUE

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: An allocation report shows where bond proceeds were assigned, while an impact report estimates environmental results.

Technical definition: Sovereign allocation evidence is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

India's FY 2023-24 report must be read with its issuance ceiling and general-revenue top-up.

MUST-WRITE KEYWORDS

- issuance
- allocation
- general revenue
- clean transport
- impact reporting

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 24:** India's FY 2023-24 report must be read with its issuance ceiling and general-revenue top-up.
- **Named evidence 24:** The report records Rs 20,000 crore of SGrB issuance and Rs 20,785.60 crore transferred to eligible schemes: Rs 16,087.99 crore clean transport, Rs 4,578.47 crore renewable energy and Rs 119.14 crore living resources/land use; Rs 785.60 crore above proceeds came from general revenue.
- **Analysis 24:** The reconciliation prevents double counting and demonstrates why source tables and accounting notes matter.
- **Qualification 24:** Budgetary transfer and modelled avoided emissions are different evidence layers.

PRELIMS TRAP

Do not sum Rs 20,785.60 crore and call it green-bond borrowing.

MAINS USE

Quote issue amount, eligible transfer, top-up source and impact status separately.

MINI RECAP

Define sovereign allocation evidence, name the evidence, trace the mechanism and retain this limit: Budgetary transfer and modelled avoided emissions are different evidence layers.

SESSION 25 - CORE - Greenium and bond pricing

VISUAL FIRST

GREEN YIELD - COMPARABLE CONVENTIONAL YIELD
|
MATCH MATURITY + LIQUIDITY + DATE
v
ESTIMATED GREENIUM, NOT IMPACT

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Greenium is the yield difference associated with a green-labelled bond relative to a suitably comparable conventional bond.

Technical definition: Greenium and bond pricing is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

A lower green yield may reduce borrowing cost, but the estimate is sensitive to the benchmark and market conditions.

MUST-WRITE KEYWORDS

- greenium
- yield spread
- benchmark bond
- liquidity
- additionality

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 25:** A lower green yield may reduce borrowing cost, but the estimate is sensitive to the benchmark and market conditions.
- **Named evidence 25:** Indian official SGrB allocation reports describe issuance and allocation but do not establish one universal greenium.
- **Analysis 25:** Differences in maturity, liquidity, tax treatment, demand and issuance timing can mimic or obscure a green premium.
- **Qualification 25:** A small or absent greenium does not prove environmental failure, and a greenium does not prove additionality.

PRELIMS TRAP

Never compare unmatched maturities and call the yield gap a greenium.

MAINS USE

Use matched-bond evidence and keep financial pricing separate from environmental impact.

MINI RECAP

Define greenium and bond pricing, name the evidence, trace the mechanism and retain this limit: A small or absent greenium does not prove environmental failure, and a greenium does not prove additionality.

SESSION 26 - CORE - Green deposits

VISUAL FIRST

DEPOSITOR -> FIXED-PERIOD DEPOSIT
BANK / NBFC -> LIABILITY
PROCEEDS -> ELIGIBLE GREEN FINANCE
REPORT -> ASSURANCE + IMPACT

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A green deposit is an interest-bearing fixed-period deposit whose proceeds are earmarked for green finance under RBI rules.

Technical definition: Green deposits is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Deposit protection and repayment terms follow the institution and deposit law; the green label governs allocation and disclosure.

MUST-WRITE KEYWORDS

- green deposit
- fixed period
- regulated entity
- financing framework
- assurance

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 26:** Deposit protection and repayment terms follow the institution and deposit law; the green label governs allocation and disclosure.
- **Named evidence 26:** RBI's 28 November 2025 directions cover commercial banks, small finance banks and specified deposit-taking NBFCs/HFCs, consolidating the earlier framework effective 1 June 2023.
- **Analysis 26:** Board policy, financing framework, external review, annual allocation assurance and impact assessment address greenwashing.
- **Qualification 26:** Pending a final Indian taxonomy, RBI's interim eligible and exclusion lists apply within the specified perimeter.

PRELIMS TRAP

A green deposit is not a mutual fund, bond or guaranteed high-return product.

MAINS USE

Separate depositor claim, bank balance-sheet liability and use-of-proceeds tracking.

MINI RECAP

Define green deposits, name the evidence, trace the mechanism and retain this limit: Pending a final Indian taxonomy, RBI's interim eligible and exclusion lists apply within the specified perimeter.

SESSION 27 - CORE - Priority-sector and transition finance

VISUAL FIRST

PSL ELIGIBILITY? -> SEPARATE TEST
GREEN TAXONOMY? -> SEPARATE TEST
TRANSITION PLAN? -> MILESTONES
NO AUTOMATIC EQUIVALENCE

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Priority-sector classification directs bank credit to specified categories, while transition finance supports credible movement of emissions-intensive activities toward lower-emission pathways.

Technical definition: Priority-sector and transition finance is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Green, priority-sector and transition labels overlap only when each framework's criteria are separately met.

MUST-WRITE KEYWORDS

- **priority sector**
- **transition finance**
- **pathway**
- **milestone**
- **lock-in**

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 27:** Green, priority-sector and transition labels overlap only when each framework's criteria are separately met.
- **Named evidence 27:** RBI's 2025 priority-sector directions recognise renewable-energy loans within stated borrower limits, while green-deposit directions permit eligible activities and SGrB investment.
- **Analysis 27:** Directed credit can address access or development gaps, whereas transition finance may fund time-bound decarbonisation plans not yet fully green.
- **Qualification 27:** Loose transition labels can extend fossil lock-in; priority-sector status does not verify climate additionality.

PRELIMS TRAP

Every renewable loan is not automatically financed by a green deposit.

MAINS USE

State the lending classification, taxonomy, transition plan, milestones and lock-in safeguards.

MINI RECAP

Define priority-sector and transition finance, name the evidence, trace the mechanism and retain this limit: Loose transition labels can extend fossil lock-in; priority-sector status does not verify climate additionality.

SESSION 28 - CORE - Physical climate risk

VISUAL FIRST

HAZARD
x EXPOSURE
x VULNERABILITY
v
ASSET / INCOME LOSS
v
FINANCIAL RISK

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Physical risk is financial loss arising from acute events such as floods and heatwaves or chronic shifts such as sea-level rise and water stress.

Technical definition: Physical climate risk is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Physical hazards affect borrowers through assets, production, income, collateral, insurance and supply chains.

MUST-WRITE KEYWORDS

- physical risk
- acute
- chronic
- exposure
- vulnerability
- collateral

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 28:** Physical hazards affect borrowers through assets, production, income, collateral, insurance and supply chains.
- **Named evidence 28:** RBI's 2024 draft disclosure framework defines acute, chronic and indirect ecosystem-service channels.
- **Analysis 28:** Repeated shocks can raise defaults, reduce collateral values, disrupt operations and concentrate losses geographically.
- **Qualification 28:** Hazard does not equal risk without exposure and vulnerability.

PRELIMS TRAP

A climate event is not automatically a bank loss of equal magnitude.

MAINS USE

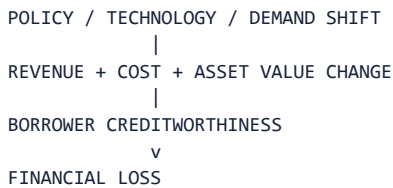
Use hazard x exposure x vulnerability, then trace to credit, market, liquidity and operational risk.

MINI RECAP

Define physical climate risk, name the evidence, trace the mechanism and retain this limit: Hazard does not equal risk without exposure and vulnerability.

SESSION 29 - CORE - Transition climate risk

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Transition risk is loss arising from policy, technology, market or preference changes during movement toward a lower-carbon economy.

Technical definition: Transition climate risk is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

An orderly early transition can reduce long-run risk even while imposing near-term repricing.

MUST-WRITE KEYWORDS

- transition risk
- policy risk
- technology risk
- repricing
- disorderly transition

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 29:** An orderly early transition can reduce long-run risk even while imposing near-term repricing.
- **Named evidence 29:** RBI identifies policy, regulation, technology and customer-preference channels in its draft climate-risk framework.
- **Analysis 29:** Carbon prices, standards or cheaper technology can impair high-emission assets and change borrower cash flows.
- **Qualification 29:** Delayed policy may create a sharper disorderly adjustment later.

PRELIMS TRAP

Transition risk is not proof that climate policy should be avoided.

MAINS USE

Compare orderly, disorderly and no-transition scenarios.

MINI RECAP

Define transition climate risk, name the evidence, trace the mechanism and retain this limit: Delayed policy may create a sharper disorderly adjustment later.

SESSION 30 - CORE - Liability risk and stranded assets

VISUAL FIRST

CLIMATE HARM / DISCLOSURE FAILURE -> CLAIM
POLICY / TECHNOLOGY SHIFT -> STRANDING
BOTH -> PROVISIONS / IMPAIRMENT / LOSS

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Liability risk arises from legal claims or compensation duties related to climate harm, disclosure or fiduciary conduct; stranded assets lose value before expected life ends.

Technical definition: Liability risk and stranded assets is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Climate risk can crystallise through courts and balance sheets as well as weather.

MUST-WRITE KEYWORDS

- liability risk
- litigation
- stranded asset
- impairment
- fiduciary duty

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 30:** Climate risk can crystallise through courts and balance sheets as well as weather.
- **Named evidence 30:** RBI speeches identify collateral and investment losses from stranded assets as a transition-risk channel.
- **Analysis 30:** A change in law, technology or demand can shorten the economic life of carbon-intensive reserves, plants or equipment.
- **Qualification 30:** Stranding depends on scenario, costs and policy; not every fossil-linked asset immediately becomes worthless.

PRELIMS TRAP

Physical, transition and liability risk overlap but are not synonyms.

MAINS USE

Trace trigger, legal or market channel, asset repricing and who bears the loss.

MINI RECAP

Define liability risk and stranded assets, name the evidence, trace the mechanism and retain this limit: Stranding depends on scenario, costs and policy; not every fossil-linked asset immediately becomes worthless.

SESSION 31 - CORE - Climate risk and financial stability

VISUAL FIRST

CLIMATE SHOCK

- > BORROWER LOSS
- > DEFAULT / COLLATERAL FALL
- > LENDER LOSS / FIRE SALE
- > SYSTEM FEEDBACK

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Climate risk becomes systemic when correlated losses, common exposures and feedback loops impair financial intermediation.

Technical definition: Climate risk and financial stability is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Microprudential resilience and macroprudential monitoring are complementary.

MUST-WRITE KEYWORDS

- financial stability
- concentration
- fire sale
- insurance gap
- feedback loop

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 31:** Microprudential resilience and macroprudential monitoring are complementary.
- **Named evidence 31:** RBI states that climate risk can transmit through credit, market and operational risk and has financial-stability implications.
- **Analysis 31:** Concentrated lending, insurance withdrawal, asset fire sales and disrupted payment or branch operations can amplify shocks.
- **Qualification 31:** Long horizons, data gaps and model uncertainty make calibration difficult.

PRELIMS TRAP

Green lending alone does not neutralise legacy portfolio exposure.

MAINS USE

Map shock -> borrower -> asset price/collateral -> institution -> system feedback.

MINI RECAP

Define climate risk and financial stability, name the evidence, trace the mechanism and retain this limit: Long horizons, data gaps and model uncertainty make calibration difficult.

SESSION 32 - CORE - RBI governance and disclosure status

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Climate-risk governance assigns board oversight, strategy, risk management, metrics and targets to regulated financial institutions.

Technical definition: RBI governance and disclosure status is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

India's final green-deposit directions and separate draft climate-disclosure framework must not be merged.

MUST-WRITE KEYWORDS

- governance
- strategy
- risk management
- metrics
- targets
- draft disclosure

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 32:** India's final green-deposit directions and separate draft climate-disclosure framework must not be merged.
- **Named evidence 32:** RBI issued its disclosure draft on 28 February 2024; a March 2025 speech said finalisation and scenario-analysis guidance were under development. No final disclosure notification was identified by 10 September 2026.
- **Analysis 32:** Structured disclosure can improve pricing and supervision, while governance embeds risk in lending and capital decisions.
- **Qualification 32:** Disclosure quality depends on data, scope assumptions and assurance and is not a substitute for prudential action.

PRELIMS TRAP

The 28 November 2025 green-deposit directions do not silently finalise the 2024 disclosure draft.

MAINS USE

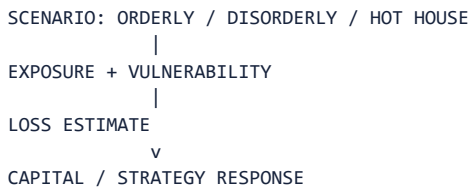
Date draft/final status and distinguish facilitative green finance from prudential climate-risk management.

MINI RECAP

Define rbi governance and disclosure status, name the evidence, trace the mechanism and retain this limit: Disclosure quality depends on data, scope assumptions and assurance and is not a substitute for prudential action.

SESSION 33 - CORE - Scenario analysis and climate stress testing

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Climate scenario analysis tests financial resilience under coherent alternative hazard and transition pathways; stress testing quantifies selected portfolio effects.

Technical definition: Scenario analysis and climate stress testing is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Scenarios reveal vulnerability rather than predict one exact future.

MUST-WRITE KEYWORDS

- scenario analysis
- stress test
- RB-CRIS
- horizon
- non-linearity

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 33:** Scenarios reveal vulnerability rather than predict one exact future.
- **Named evidence 33:** RBI's March 2025 speech stated that guidance on climate scenario analysis and stress testing was being developed and that RB-CRIS was intended to address data gaps.
- **Analysis 33:** Long horizons and non-linear shocks require sector, geography and counterparty mapping beyond historical-loss models.
- **Qualification 33:** Results depend on assumptions, data and balance-sheet responses and should not be read as forecasts.

PRELIMS TRAP

A severe scenario is not RBI's prediction of what will happen.

MAINS USE

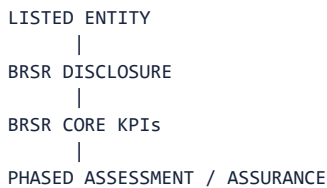
Disclose scenario, horizon, transmission channel, assumptions and management response.

MINI RECAP

Define scenario analysis and climate stress testing, name the evidence, trace the mechanism and retain this limit: Results depend on assumptions, data and balance-sheet responses and should not be read as forecasts.

SESSION 34 - CORE - SEBI BRSR and BRSR Core

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: BRSR is SEBI's sustainability-reporting framework for covered listed entities; BRSR Core is a subset of key indicators subject to phased assessment or assurance.

Technical definition: SEBI BRSR and BRSR Core is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Disclosure standardisation improves comparability but does not itself prove sustainable performance.

MUST-WRITE KEYWORDS

- BRSR
- BRSR Core
- ESG
- listed entity
- assessment
- value chain

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 34:** Disclosure standardisation improves comparability but does not itself prove sustainable performance.
- **Named evidence 34:** SEBI's 28 March 2025 circular sets the BRSR Core glide path at top 500 listed entities for FY 2025-26 and top 1,000 for FY 2026-27; value-chain disclosure is voluntary from FY 2025-26.
- **Analysis 34:** Common metrics can improve investor discipline and reveal environmental and social dependencies.
- **Qualification 34:** Entity boundaries, estimates and value-chain data quality constrain comparability.

PRELIMS TRAP

RBI does not mandate BRSR; SEBI does within the listed-company perimeter.

MAINS USE

Separate ordinary BRSR coverage, BRSR Core assessment/assurance and voluntary value-chain disclosure.

MINI RECAP

Define sebi brsr and brsr core, name the evidence, trace the mechanism and retain this limit: Entity boundaries, estimates and value-chain data quality constrain comparability.

SESSION 35 - CORE - SEBI green debt and transition bonds

VISUAL FIRST

ISSUER CREDIT RISK
+
LABEL / USE-OF-PROCEEDS RULES
+
TRACKING / REVIEW
v
GREEN OR TRANSITION DEBT

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: SEBI-regulated green debt raises funds for eligible environmental projects; transition bonds fund disclosed movement toward lower-emission operations.

Technical definition: SEBI green debt and transition bonds is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Use-of-proceeds discipline, reporting and anti-greenwashing duties distinguish labelled debt from ordinary borrowing.

MUST-WRITE KEYWORDS

- green debt security
- transition bond
- blue bond
- yellow bond
- disclosure

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 35:** Use-of-proceeds discipline, reporting and anti-greenwashing duties distinguish labelled debt from ordinary borrowing.
- **Named evidence 35:** SEBI issued anti-greenwashing and revised disclosure circulars on 3 and 6 February 2023, followed by transition-bond requirements on 4 May 2023.
- **Analysis 35:** Objectives, project selection, tracking, utilisation and impact information let investors test label credibility.
- **Qualification 35:** A transition bond needs a credible pathway and cannot become a permanent label for marginal improvement.

PRELIMS TRAP

Blue, yellow, green and transition labels do not eliminate issuer credit risk.

MAINS USE

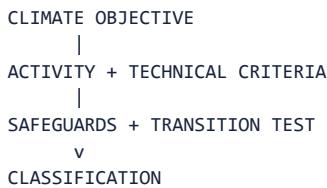
Assess eligibility, exclusions, transition plan, allocation, review and outcome.

MINI RECAP

Define sebi green debt and transition bonds, name the evidence, trace the mechanism and retain this limit: A transition bond needs a credible pathway and cannot become a permanent label for marginal improvement.

SESSION 36 - CORE - Climate taxonomy

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A climate-finance taxonomy classifies activities or measures according to climate objectives and specified criteria.

Technical definition: Climate taxonomy is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Taxonomy reduces information asymmetry only when thresholds, transition pathways and safeguards are clear.

MUST-WRITE KEYWORDS

- taxonomy
- threshold
- transition pathway
- interoperability
- do no significant harm

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 36:** Taxonomy reduces information asymmetry only when thresholds, transition pathways and safeguards are clear.
- **Named evidence 36:** DEA released a draft framework on 7 May 2025 and invited comments to 25 June 2025; no final national taxonomy was identified by 10 September 2026.
- **Analysis 36:** Sector annexures can guide capital allocation and comparability while recognising India's development and energy-access context.
- **Qualification 36:** Rigid binary labels may exclude credible transition activity or encourage threshold gaming.

PRELIMS TRAP

A draft framework is not a final binding taxonomy.

MAINS USE

State objective, activity criteria, threshold, do-no-significant-harm safeguard and transition treatment.

MINI RECAP

Define climate taxonomy, name the evidence, trace the mechanism and retain this limit: Rigid binary labels may exclude credible transition activity or encourage threshold gaming.

SESSION 37 - CORE - Greenwashing and verification

VISUAL FIRST

LABEL

- > ELIGIBILITY
- > ALLOCATION
- > OUTPUT
- > OUTCOME
- > ADDITIONALITY

BREAK ANY LINK -> GREENWASHING RISK

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Greenwashing is misleading environmental labelling or selective disclosure that overstates alignment, allocation or impact.

Technical definition: Greenwashing and verification is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Credibility requires a chain from taxonomy to proceeds, outputs, outcomes and additionality.

MUST-WRITE KEYWORDS

- greenwashing
- taxonomy
- allocation
- assurance
- impact
- additionality

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 37:** Credibility requires a chain from taxonomy to proceeds, outputs, outcomes and additionality.
- **Named evidence 37:** RBI defines greenwashing in its green-deposit directions, while SEBI's February 2023 circular specifies issuer dos and don'ts for green debt.
- **Analysis 37:** External review, allocation assurance and impact reporting reduce information asymmetry and reputational risk.
- **Qualification 37:** Verification can test stated criteria but cannot guarantee every modelled outcome or eliminate conflicts.

PRELIMS TRAP

A second-party opinion is not a sovereign or regulator guarantee.

MAINS USE

Use the integrity chain: label -> eligibility -> allocation -> output -> outcome -> additionality.

MINI RECAP

Define greenwashing and verification, name the evidence, trace the mechanism and retain this limit: Verification can test stated criteria but cannot guarantee every modelled outcome or eliminate conflicts.

SESSION 38 - CORE - Just transition

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A just transition manages the worker, community, consumer and regional consequences of decarbonisation.

Technical definition: Just transition is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Political durability depends on distributing both transition costs and new opportunities fairly.

MUST-WRITE KEYWORDS

- just transition
- distribution
- workers
- regions
- affordability
- participation

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 38:** Political durability depends on distributing both transition costs and new opportunities fairly.
- **Named evidence 38:** Coal-dependent districts, informal workers and low-income energy users face distinct adjustment risks.
- **Analysis 38:** Skills, income support, social protection, regional diversification, affordable energy and worker participation can convert resistance into credible transition.
- **Qualification 38:** Compensation alone is insufficient without alternative productive capacity and local public services.

PRELIMS TRAP

Just transition does not mean indefinite protection of every high-emission activity.

MAINS USE

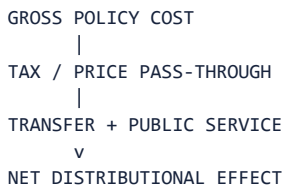
Identify losers, timing, place, fiscal instrument, participation and new livelihood pathway.

MINI RECAP

Define just transition, name the evidence, trace the mechanism and retain this limit: Compensation alone is insufficient without alternative productive capacity and local public services.

SESSION 39 - CORE - Distributional incidence of climate policy

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Distributional incidence identifies who bears policy costs and who receives benefits across income groups, places and generations.

Technical definition: Distributional incidence of climate policy is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

A policy can be efficient in aggregate yet inequitable in incidence.

MUST-WRITE KEYWORDS

- incidence
- pass-through
- compensation
- lifeline
- progressive recycling

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 39:** A policy can be efficient in aggregate yet inequitable in incidence.
- **Named evidence 39:** Carbon prices may pass through to energy and transport costs, while clean-air and climate benefits may accrue differently across groups.
- **Analysis 39:** Targeted transfers, lifeline consumption, public transport and progressive recycling can protect vulnerable users without erasing the signal.
- **Qualification 39:** Universal subsidies can be fiscally costly and disproportionately benefit high consumers.

PRELIMS TRAP

Average household effects hide rural, gender, occupational and regional differences.

MAINS USE

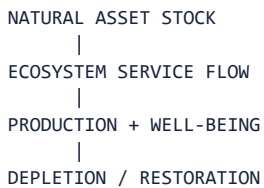
Present gross burden, compensation, net incidence and long-term benefit separately.

MINI RECAP

Define distributional incidence of climate policy, name the evidence, trace the mechanism and retain this limit: Universal subsidies can be fiscally costly and disproportionately benefit high consumers.

SESSION 40 - CORE - Natural capital

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Natural capital is the stock of ecosystems and natural resources that generates services supporting production and welfare.

Technical definition: Natural capital is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Economic growth can deplete natural assets even when measured GDP rises.

MUST-WRITE KEYWORDS

- natural capital
- ecosystem services
- depletion
- degradation
- asset account

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 40:** Economic growth can deplete natural assets even when measured GDP rises.
- **Named evidence 40:** Forests, soils, water bodies and biodiversity provide provisioning, regulating, cultural and supporting services.
- **Analysis 40:** Asset accounts reveal depletion and degradation that ordinary production flows may omit.
- **Qualification 40:** Monetary valuation is incomplete for ecological thresholds, non-market values and irreversible loss.

PRELIMS TRAP

Nature is not fully substitutable by produced capital merely because a price is assigned.

MAINS USE

Use natural-capital accounts as a satellite lens alongside, not a mechanical replacement for, GDP.

MINI RECAP

Define natural capital, name the evidence, trace the mechanism and retain this limit: Monetary valuation is incomplete for ecological thresholds, non-market values and irreversible loss.

SESSION 41 - CORE - Environmental accounting and green GDP

VISUAL FIRST

NATIONAL ACCOUNTS + PHYSICAL ENVIRONMENT DATA
|
ASSET / FLOW / ECOSYSTEM ACCOUNTS
v
SUSTAINABILITY DASHBOARD

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Environmental-economic accounting links physical environmental stocks and flows with economic accounts; green-GDP adjustments attempt to reflect depletion and degradation.

Technical definition: Environmental accounting and green GDP is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

A single adjusted number cannot replace a dashboard of physical and monetary accounts.

MUST-WRITE KEYWORDS

- SEEA
- environmental account
- ecosystem account
- depletion
- green GDP

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 41:** A single adjusted number cannot replace a dashboard of physical and monetary accounts.
- **Named evidence 41:** MoSPI's EnviStats India 2025 follows the System of Environmental-Economic Accounting and includes asset and ecosystem accounts.
- **Analysis 41:** Linking land, water, energy, emissions and ecosystem services to economic activity improves sustainability analysis.
- **Qualification 41:** Valuation methods, data gaps and non-market ecological functions limit comparability.

PRELIMS TRAP

Green GDP is not simply GDP minus all environmental expenditure.

MAINS USE

Cross-link Topic 1: distinguish production flow, asset stock, defensive expenditure and welfare.

MINI RECAP

Define environmental accounting and green gdp, name the evidence, trace the mechanism and retain this limit: Valuation methods, data gaps and non-market ecological functions limit comparability.

SESSION 42 - CORE - Circular economy hierarchy

VISUAL FIRST

REFUSE / REDESIGN

↓

REDUCE → REUSE → REPAIR

↓

REFURBISH → REMANUFACTURE

↓

RECYCLE → RECOVER → DISPOSE

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A circular economy designs out waste, keeps products and materials at highest value and regenerates natural systems.

Technical definition: Circular economy hierarchy is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Prevention and life extension usually preserve more embedded value than material recycling.

MUST-WRITE KEYWORDS

- refuse
- reduce
- reuse
- repair
- refurbish
- remanufacture
- recycle

CLAIM → NAMED EVIDENCE → ANALYSIS → QUALIFICATION

- **Claim 42:** Prevention and life extension usually preserve more embedded value than material recycling.
- **Named evidence 42:** The 2025 UPSC official key recognises lower raw-material use and lower production waste as channels reducing greenhouse-gas emissions.
- **Analysis 42:** Refuse, reduce, redesign, reuse, repair, refurbish and remanufacture precede recycling and recovery.
- **Qualification 42:** Material loops still require energy and can create pollution or rebound.

PRELIMS TRAP

Recycling alone is not a circular economy.

MAINS USE

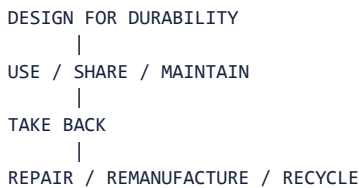
Use the hierarchy and explain why value retention declines down the loop.

MINI RECAP

Define circular economy hierarchy, name the evidence, trace the mechanism and retain this limit: Material loops still require energy and can create pollution or rebound.

SESSION 43 - CORE - Circular business models

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Circular business models extend product life or recover value through sharing, leasing, product-as-a-service, repair, refurbishment and remanufacturing.

Technical definition: Circular business models is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Ownership and revenue design can align producers with durability and lifecycle cost.

MUST-WRITE KEYWORDS

- product as a service
- sharing
- reverse logistics
- durability
- remanufacture

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 43:** Ownership and revenue design can align producers with durability and lifecycle cost.
- **Named evidence 43:** Service contracts can reward uptime and resource efficiency instead of one-time sales.
- **Analysis 43:** Reverse logistics and secondary-material markets convert end-of-life products into inputs.
- **Qualification 43:** Sharing can increase utilisation but may also increase total use or shift labour risks.

PRELIMS TRAP

A rental platform is not circular unless lifecycle resource use actually falls.

MAINS USE

Test design, utilisation, lifetime, reverse flow and net material impact.

MINI RECAP

Define circular business models, name the evidence, trace the mechanism and retain this limit: Sharing can increase utilisation but may also increase total use or shift labour risks.

SESSION 44 - CORE - Life-cycle assessment

VISUAL FIRST

RAW MATERIAL -> MANUFACTURE -> TRANSPORT
-> USE -> REPAIR / REUSE
-> RECYCLE / DISPOSAL
MEASURE PER FUNCTIONAL UNIT

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Life-cycle assessment evaluates environmental impacts across extraction, production, transport, use and end of life.

Technical definition: Life-cycle assessment is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

A technology can shift rather than eliminate impacts across stages or locations.

MUST-WRITE KEYWORDS

- LCA
- functional unit
- system boundary
- embodied emissions
- end of life

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 44:** A technology can shift rather than eliminate impacts across stages or locations.
- **Named evidence 44:** LCA uses a defined functional unit, system boundary, inventory and impact assessment.
- **Analysis 44:** It exposes embodied emissions, water use, material extraction and disposal impacts hidden by tailpipe or factory-only metrics.
- **Qualification 44:** Results depend on electricity mix, lifetime, recycling assumptions and allocation method.

PRELIMS TRAP

Zero operational emissions does not mean zero lifecycle impact.

MAINS USE

State functional unit, boundary, data year and sensitivity before comparing alternatives.

MINI RECAP

Define life-cycle assessment, name the evidence, trace the mechanism and retain this limit: Results depend on electricity mix, lifetime, recycling assumptions and allocation method.

SESSION 45 - CORE - Resource efficiency and material productivity

VISUAL FIRST

VALUE ADDED / MATERIAL INPUT
= MATERIAL PRODUCTIVITY
CHECK ALSO:
TOTAL EXTRACTION + TOTAL WASTE

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Resource efficiency produces more welfare or value with less material, energy, water and waste over the lifecycle.

Technical definition: Resource efficiency and material productivity is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Material productivity complements labour and capital productivity in green industrial strategy.

MUST-WRITE KEYWORDS

- resource efficiency
- material productivity
- throughput
- intensity
- absolute use

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 45:** Material productivity complements labour and capital productivity in green industrial strategy.
- **Named evidence 45:** NITI Aayog's Circular Economy Cell coordinates sectoral action plans and resource-efficiency work.
- **Analysis 45:** Light-weighting, design standards, secondary inputs and process integration can reduce import dependence and pollution.
- **Qualification 45:** Ratios can improve while total material use rises if output grows faster.

PRELIMS TRAP

Efficiency is not sufficient when scale effects overwhelm savings.

MAINS USE

Report both intensity and absolute throughput.

MINI RECAP

Define resource efficiency and material productivity, name the evidence, trace the mechanism and retain this limit: Ratios can improve while total material use rises if output grows faster.

SESSION 46 - CORE - Rebound effect

VISUAL FIRST

EFFICIENCY -> LOWER UNIT COST
|
MORE USE / SPENDING
|
PART OF EXPECTED SAVING OFFSET
v
MEASURE ABSOLUTE OUTCOME

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: The rebound effect occurs when lower unit cost from efficiency induces additional use that offsets part of the expected resource saving.

Technical definition: Rebound effect is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Efficiency gains require absolute-outcome monitoring.

MUST-WRITE KEYWORDS

- rebound
- direct
- indirect
- economy-wide
- absolute consumption

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 46:** Efficiency gains require absolute-outcome monitoring.
- **Named evidence 46:** Cheaper energy services, transport or material use can increase demand after an efficiency improvement.
- **Analysis 46:** Direct rebound affects the same service; indirect rebound reallocates savings; economy-wide rebound changes production patterns.
- **Qualification 46:** Rebound rarely means efficiency is useless, but it reduces expected savings.

PRELIMS TRAP

An efficiency percentage cannot be treated as an equal fall in total emissions.

MAINS USE

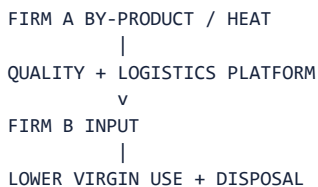
Combine efficiency standards with price signals and absolute targets where necessary.

MINI RECAP

Define rebound effect, name the evidence, trace the mechanism and retain this limit: Rebound rarely means efficiency is useless, but it reduces expected savings.

SESSION 47 - APPLIED - Industrial symbiosis

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Industrial symbiosis links firms so one process's by-product, heat, water or material becomes another process's input.

Technical definition: Industrial symbiosis is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Geographic coordination can convert waste-disposal cost into productive resource value.

MUST-WRITE KEYWORDS

- industrial symbiosis
- by-product
- waste heat
- cluster
- secondary material

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 47:** Geographic coordination can convert waste-disposal cost into productive resource value.
- **Named evidence 47:** Industrial clusters can share utilities, recovery facilities, logistics and material-information systems.
- **Analysis 47:** Dense networks lower transaction and transport costs and support consistent secondary-material supply.
- **Qualification 47:** Contamination, liability, quality standards and dependence on one supplier can limit exchanges.

PRELIMS TRAP

Using waste as an input does not remove the need to prevent hazardous generation.

MAINS USE

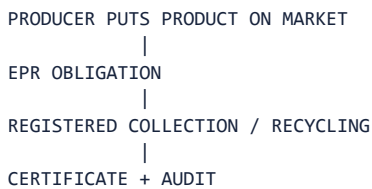
Map material flow, quality specification, liability and net lifecycle benefit.

MINI RECAP

Define industrial symbiosis, name the evidence, trace the mechanism and retain this limit: Contamination, liability, quality standards and dependence on one supplier can limit exchanges.

SESSION 48 - APPLIED - Extended Producer Responsibility

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Extended Producer Responsibility makes producers responsible for specified post-consumer collection, recycling or environmentally sound management.

Technical definition: Extended Producer Responsibility is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

EPR shifts lifecycle incentives upstream but requires credible measurement and enforcement downstream.

MUST-WRITE KEYWORDS

- EPR
- producer
- collection
- recycler
- certificate
- environmental compensation

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 48:** EPR shifts lifecycle incentives upstream but requires credible measurement and enforcement downstream.
- **Named evidence 48:** Indian rules use registration, targets, authorised recyclers, EPR certificates, returns and environmental compensation across specified waste streams.
- **Analysis 48:** A producer facing end-of-life obligations has stronger incentives for design, take-back and recycled content.
- **Qualification 48:** Certificate systems can suffer fake recycling, double counting and exclusion of informal workers.

PRELIMS TRAP

EPR does not mean the producer must physically collect every item itself.

MAINS USE

Trace producer obligation -> registered processor -> verified certificate -> target adjustment.

MINI RECAP

Define extended producer responsibility, name the evidence, trace the mechanism and retain this limit: Certificate systems can suffer fake recycling, double counting and exclusion of informal workers.

SESSION 49 - APPLIED - India's dated EPR rule map

VISUAL FIRST

PLASTIC | BATTERY | E-WASTE | TYRE | USED OIL
EACH HAS SEPARATE RULE
PRODUCER -> PORTAL -> PROCESSOR -> CERTIFICATE

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: India applies separate EPR regimes to different waste streams rather than one universal recycling rule.

Technical definition: India's dated EPR rule map is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Each product category has its own definitions, targets, portal and authorised processing route.

MUST-WRITE KEYWORDS

- plastic packaging
- battery
- e-waste
- tyre
- used oil
- CPCB

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 49:** Each product category has its own definitions, targets, portal and authorised processing route.
- **Named evidence 49:** Plastic-packaging EPR guidelines were notified in February 2022; Battery Waste Management Rules on 22 August 2022; waste-tyre EPR on 21 July 2022; E-Waste Rules on 2 November 2022 effective 1 April 2023; and used-oil EPR through the 18 September 2023 amendment.
- **Analysis 49:** Category-specific rules recognise distinct material, safety and recycling chains.
- **Qualification 49:** Targets and amendments change over time and should be checked in the current notification and CPCB portal.

PRELIMS TRAP

Registration under one EPR regime does not satisfy another waste category.

MAINS USE

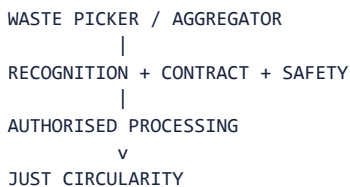
Name waste stream, rule date, obligated actor, target and approved processor.

MINI RECAP

Define india's dated epr rule map, name the evidence, trace the mechanism and retain this limit: Targets and amendments change over time and should be checked in the current notification and CPCB portal.

SESSION 50 - APPLIED - Informal sector and circularity

VISUAL FIRST



Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Informal collectors and processors recover significant material but often lack secure livelihoods, safety and recognition.

Technical definition: Informal sector and circularity is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

A just circular transition must formalise rights and standards without destroying collection networks.

MUST-WRITE KEYWORDS

- informal worker
- waste picker
- occupational safety
- integration
- traceability

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 50:** A just circular transition must formalise rights and standards without destroying collection networks.
- **Named evidence 50:** Indian waste systems commonly depend on waste pickers, aggregators and small recyclers alongside municipal and producer systems.
- **Analysis 50:** Contracts, cooperatives, occupational safety, traceable payments and access to material can improve both recovery and dignity.
- **Qualification 50:** Abrupt exclusion can reduce livelihoods and even collection efficiency.

PRELIMS TRAP

Formalisation is not merely app registration or displacement by large firms.

MAINS USE

Include worker integration, safety, bargaining power and verified processing in EPR evaluation.

MINI RECAP

Define informal sector and circularity, name the evidence, trace the mechanism and retain this limit: Abrupt exclusion can reduce livelihoods and even collection efficiency.

SESSION 51 - APPLIED - Climate clubs and CBAM

VISUAL FIRST

DOMESTIC CARBON CONSTRAINT
|
IMPORT EMBEDDED EMISSIONS
|
BORDER ADJUSTMENT
v
LEAKAGE / EQUITY / TRADE DEBATE

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A climate club coordinates stronger climate action among participants, while a carbon border adjustment charges specified imports for embedded emissions under importing-jurisdiction rules.

Technical definition: Climate clubs and CBAM is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Border measures link climate ambition with competitiveness but raise equity, measurement and trade-law concerns.

MUST-WRITE KEYWORDS

- climate club
- CBAM
- embedded emissions
- leakage
- competitiveness
- MRV

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 51:** Border measures link climate ambition with competitiveness but raise equity, measurement and trade-law concerns.
- **Named evidence 51:** The EU CBAM definitive regime began on 1 January 2026 for specified goods including cement, iron and steel, aluminium, fertilisers, electricity and hydrogen.
- **Analysis 51:** Embedded-emissions MRV and recognition of carbon prices paid abroad shape exporter incentives.
- **Qualification 51:** CBAM is an EU measure, not part of India's CCTS; recognition of Indian carbon costs cannot be assumed.

PRELIMS TRAP

A border adjustment is not identical to an ordinary customs tariff.

MAINS USE

Cross-link Topic 20 for WTO law; here focus on decarbonisation incentives and transition finance.

MINI RECAP

Define climate clubs and cbam, name the evidence, trace the mechanism and retain this limit: CBAM is an EU measure, not part of India's CCTS; recognition of Indian carbon costs cannot be assumed.

SESSION 52 - APPLIED - Innovation and green public investment

VISUAL FIRST

CARBON SIGNAL -> DEPLOYMENT
R&D SUPPORT -> NEW OPTIONS
PUBLIC INVESTMENT -> NETWORKS
SOCIAL POLICY -> LEGITIMACY

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Green innovation policy supports knowledge creation and deployment where private investors cannot capture all learning benefits.

Technical definition: Innovation and green public investment is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Carbon pricing encourages adoption, while research support and public infrastructure expand the available technology set.

MUST-WRITE KEYWORDS

- innovation spillover
- demonstration
- network infrastructure
- learning curve
- sunset

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 52:** Carbon pricing encourages adoption, while research support and public infrastructure expand the available technology set.
- **Named evidence 52:** Grid, storage, public transport, resilient water systems and research platforms have network or public-good features.
- **Analysis 52:** Mission-oriented procurement, grants and demonstration finance can reduce learning costs and coordinate complementary assets.
- **Qualification 52:** Poor technology selection can create lock-in, fiscal waste or stranded public assets.

PRELIMS TRAP

A subsidy or installed capacity figure does not establish cost-effective additional emissions reduction.

MAINS USE

Evaluate innovation spillover, infrastructure complement, competition, sunset and measured outcome.

MINI RECAP

Define innovation and green public investment, name the evidence, trace the mechanism and retain this limit: Poor technology selection can create lock-in, fiscal waste or stranded public assets.

SESSION 53 - APPLIED - Climate and circular policy dashboard

VISUAL FIRST

INPUT -> FINANCE / RULE

OUTPUT -> ASSET / CERTIFICATE

OUTCOME -> EMISSIONS / RESILIENCE

IMPACT -> WELFARE + DISTRIBUTION

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: A policy dashboard evaluates climate action through inputs, outputs, outcomes, resilience, distribution and fiscal risk.

Technical definition: Climate and circular policy dashboard is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Headline finance or certificate counts are incomplete without environmental and social results.

MUST-WRITE KEYWORDS

- input
- output
- outcome
- additionality
- resilience
- distribution
- fiscal risk

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 53:** Headline finance or certificate counts are incomplete without environmental and social results.
- **Named evidence 53:** Official frameworks separately report issuance, allocation, impact, disclosure coverage and compliance status.
- **Analysis 53:** Good evaluation combines absolute emissions, intensity, avoided loss, material throughput, service quality, jobs, affordability and verified additionality.
- **Qualification 53:** Attribution is difficult because technology, energy prices, growth and other policies change simultaneously.

PRELIMS TRAP

Do not infer avoided emissions from money allocated or registrations completed.

MAINS USE

End Mains answers with a measurable dashboard and review mechanism.

MINI RECAP

Define climate and circular policy dashboard, name the evidence, trace the mechanism and retain this limit: Attribution is difficult because technology, energy prices, growth and other policies change simultaneously.

SESSION 54 - APPLIED - Integrated regulatory architecture

VISUAL FIRST

MOEFCC -> ENVIRONMENT RULES
BEE / CERC -> CERTIFICATES / TRADING
DEA -> SGrB / TAXONOMY
RBI / SEBI -> FINANCE / DISCLOSURE
CPCB / STATES -> IMPLEMENTATION

Reading rule: move left to right unless an arrow or hierarchy shows otherwise.

DEFINITION

Plain-language definition: Climate governance distributes functions across environmental, energy, financial, securities, statistical and local authorities.

Technical definition: Integrated regulatory architecture is analysed through its legal or accounting boundary, causal mechanism, measurement unit, implementation stage, distributional effect and stated qualification.

ANSWER-GRABBING LINE

Regulation, disclosure, finance, verification and implementation must be assigned to the correct institution.

MUST-WRITE KEYWORDS

- MoEFCC
- BEE
- CERC
- DEA
- RBI
- SEBI
- CPCB

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim 54:** Regulation, disclosure, finance, verification and implementation must be assigned to the correct institution.
- **Named evidence 54:** MoEFCC sets environmental rules; BEE administers CCTS and PAT; CERC oversees certificate trading; DEA owns sovereign green-bond and taxonomy frameworks; RBI regulates covered lenders and deposits; SEBI regulates listed disclosure and green debt; CPCB and State Boards support waste-rule implementation.
- **Analysis 54:** Institutional separation creates specialisation and checks but requires data interoperability and coordination.
- **Qualification 54:** No regulator is a supranational climate government, and disclosure cannot replace environmental enforcement.

PRELIMS TRAP

Do not attribute BRSR to RBI or CCTS administration to SEBI.

MAINS USE

Use an authority-function-instrument-status table in Mains answers.

MINI RECAP

Define integrated regulatory architecture, name the evidence, trace the mechanism and retain this limit: No regulator is a supranational climate government, and disclosure cannot replace environmental enforcement.

PYQS AND ANSWER PRACTICE

VERIFIED MAINS PYQ 1 - 2025 GS-III - 10 MARKS

Question: How can India achieve energy independence through clean technology by 2047? How can biotechnology play a crucial role in this endeavour? Answer in 150 words.

Model solution: Energy independence means reducing exposure to imported fossil fuels and concentrated clean-technology supply chains while preserving reliable, affordable energy. India needs renewable power, storage, stronger grids, electrified transport, efficient buildings and industry, green hydrogen for hard-to-abate uses, domestic critical-component capability and diversified mineral partnerships. Demand management and circular recovery of batteries and metals reduce import intensity. Biotechnology supports second-generation biofuels from residues, compressed biogas, algal fuels, engineered enzymes, waste-to-energy and climate-resilient biomass. It can also improve process efficiency and substitute fossil feedstocks in chemicals. Public research, demonstration finance, standards and procurement should support learning without protecting inefficient firms indefinitely. Land, water, food-security, biosafety and lifecycle-emission safeguards are essential. Energy independence is therefore diversified resilience, not autarky or a mere renewable-capacity target.

VERIFIED MAINS PYQ 2 - 2025 GS-III - 10 MARKS

Question: What is Carbon Capture, Utilization and Storage (CCUS)? What is the potential role of CCUS in tackling climate change? Answer in 150 words.

Model solution: CCUS separates carbon dioxide from concentrated industrial or energy streams, compresses and transports it, uses it in products or processes where viable, or injects it into suitable geological formations for long-term storage. It can reduce residual emissions in cement, steel, chemicals and refineries where process emissions or high-temperature heat are difficult to eliminate. It may also support low-carbon hydrogen and, when paired with sustainable biomass or direct air capture, carbon removal. However, capture consumes energy, raises cost and requires transport networks, storage characterisation, monitoring and liability rules. Utilisation may delay rather than permanently avoid release, and enhanced fossil extraction can undermine benefits. CCUS should therefore complement efficiency, electrification, renewable energy and material substitution, not preserve avoidable fossil dependence.

VERIFIED MAINS PYQ 3 - 2025 GS-III - 15 MARKS

Question: Write a review on India's climate commitments under the Paris Agreement (2015) and mention how these have been further strengthened in COP26 (2021). In this direction, how has the first Nationally Determined Contribution intended by India been updated in 2022? Answer in 250 words.

Model solution: India's first NDC under the Paris Agreement combined development with differentiated responsibility. Its 2030 goals included reducing GDP emissions intensity by 33-35 per cent from 2005, achieving 40 per cent cumulative installed electric-power capacity from non-fossil sources with support, and creating an additional 2.5-3 billion tonnes of CO₂-equivalent forest and tree carbon sink. At COP26 in 2021, India announced the Panchamrit political package: 500 GW non-fossil capacity, about 50 per cent energy requirements from renewables, one billion tonnes reduction in projected emissions to 2030, 45 per cent emissions-intensity reduction, and net zero by 2070. The formal 2022 NDC update incorporated a 45 per cent emissions-intensity reduction and about 50 per cent cumulative installed electric-power capacity from non-fossil sources by 2030, along with LiFE. Political announcements and formally communicated NDC text should remain distinct. Delivery requires grids, storage, finance, industrial decarbonisation, adaptation and a just transition. Progress should be assessed against dated official inventories and capacity data, not promotional claims.

VERIFIED OBJECTIVE PYQ 1 - 2025 PRELIMS GS-I Q4

Question: Consider the following statements: I. The Reserve Bank of India mandates all the listed companies in India to submit a Business Responsibility and Sustainability Report (BRSR). II. In India, a company submitting a BRSR makes disclosures in the report that are largely non-financial in nature.

Which of the statements given above is/are correct?

A. I only B. II only C. Both I and II D. Neither I nor II

Official Set-A key: B.

Solution: Statement I is incorrect because SEBI, not RBI, mandates BRSR within the listed-company perimeter. Statement II is correct because BRSR principally reports environmental, social and governance information. Do not confuse ordinary BRSR coverage with the phased BRSR Core assessment/assurance glide path.

VERIFIED OBJECTIVE PYQ 2 - 2025 PRELIMS GS-I Q9

Question: Consider the following statements: I. Circular economy reduces the emissions of greenhouse gases. II. Circular economy reduces the use of raw materials as inputs. III. Circular economy reduces wastage in the production process.

Which one of the following is correct in respect of the above statements?

A. Both Statement II and Statement III are correct and both of them explain Statement I B. Both Statement II and Statement III are correct but only one of them explains Statement I C. Only one of the Statements II and III is correct and that explains Statement I D. Neither Statement II nor Statement III is correct

Official Set-A key: A.

Solution: Lower virgin-material extraction and lower production waste can both reduce energy use and greenhouse-gas emissions, so Statements II and III are correct and both explain Statement I. The conclusion is a causal mechanism, not a claim that every recycling process has zero lifecycle emissions.

KEY-NEUTRAL OBJECTIVE ROUTES

- **2020 Prelims Q85:** Social Cost of Carbon as monetary valuation of incremental emissions damage. The official key is not held locally, so no answer letter is inferred.
- **2022 Prelims Q80:** Greenwashing as false or misleading environmental claims. The official key is not held locally, so no answer letter is inferred.
- **2026 Prelims Q92:** Sustainability bonds and combined environmental/social financing. The locally held key is provisional, so no answer letter is asserted.

OPTIONAL ADVANCED

ADVANCED 1 - Discounting under deep uncertainty

CORE QUESTION -> DISCOUNTING UNDER DEEP UNCERTAINTY -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Use declining or scenario-dependent discount rates only as an analytical refinement. Long horizons, fat-tail damages and irreversible thresholds weaken mechanical reliance on one constant rate, but uncertainty must not be converted into an invented SCC.

ADVANCED 2 - Prices versus quantities

CORE QUESTION -> PRICES VERSUS QUANTITIES -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Under uncertain abatement costs and damages, tax and cap choices trade price certainty against quantity certainty. Hybrid designs use price floors, ceilings, reserves or adjustment corridors, but administrative simplicity and credibility remain central.

ADVANCED 3 - Carbon leakage and consumption accounting

CORE QUESTION -> CARBON LEAKAGE AND CONSUMPTION ACCOUNTING -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Territorial inventories allocate emissions where production occurs; consumption-based accounts reassign embodied emissions to final demand. Leakage claims require a counterfactual showing relocation or import substitution caused by policy.

ADVANCED 4 - Financed emissions

CORE QUESTION -> FINANCED EMISSIONS -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Financed emissions attribute part of a borrower's or investee's emissions to loans and investments. Portfolio totals depend on data coverage, attribution factors and Scope 1-3 boundaries; they are risk and target tools, not bank-owned physical emissions.

ADVANCED 5 - Double materiality

CORE QUESTION -> DOUBLE MATERIALITY -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Financial materiality asks how climate affects enterprise value; impact materiality asks how the entity affects people and environment. The two perspectives can overlap but should not be silently treated as the same disclosure test.

ADVANCED 6 - Adaptation metrics

CORE QUESTION -> ADAPTATION METRICS -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Mitigation often uses tonnes of CO₂-equivalent; adaptation requires context-specific metrics such as people protected, service continuity, avoided losses or reduced vulnerability. A universal single adaptation unit remains elusive.

ADVANCED 7 - Circular rebound and trade-offs

CORE QUESTION -> CIRCULAR REBOUND AND TRADE-OFFS -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Longer product life and secondary materials can reduce throughput, but cheap access may increase use and recycling may shift pollution. Measure absolute virgin input, energy, toxicity and service delivered over the full lifecycle.

ADVANCED 8 - Dynamic transition finance

CORE QUESTION -> DYNAMIC TRANSITION FINANCE -> MEASUREMENT / INCENTIVE / DISTRIBUTION TEST

Transition finance should be conditional on science-aligned pathways, capital expenditure, interim milestones and consequences for failure. A static label without declining emissions intensity, absolute-emissions checks and lock-in safeguards is weak.

CONSOLIDATED REGISTER NOTES

1. Core economic problem

- Climate change is a negative externality, a global public bad and a collective-action problem.
- The atmosphere's limited absorptive capacity has commons-like features.
- Long-lived greenhouse gases make climate policy intergenerational.
- Private cost can remain below social cost unless policy changes incentives.
- Uncertainty strengthens the need for transparent scenarios; it does not make expected damage zero.

2. Response taxonomy

Response	Core function	Typical finance	Boundary
Mitigation	Reduce emissions or enhance sinks	Clean energy, efficiency, industrial change	Limits future hazard
Adaptation	Reduce exposure or vulnerability	Resilient infrastructure, water, health, crops	Manages impacts
Loss and damage	Address residual economic/non-economic harm	Grants and dedicated international arrangements	Not a substitute for either mitigation or adaptation

3. Carbon-value and market distinctions

- SCC = discounted model-based marginal damage per additional tonne of CO₂-equivalent.
- Carbon tax = government-set price; emissions quantity responds.
- Cap-and-trade = regulator-set aggregate allowance quantity; market price responds.
- Baseline-and-credit = credits/shortfalls relative to a benchmark, often intensity-linked.
- Offset = project reduction/removal relative to a counterfactual baseline.
- Integrity tests: baseline, additionality, leakage, permanence, MRV, registry and no double counting.

4. Indian certificate map

Instrument	What it represents	Institution / legal rail	Trap
ESCert	Verified energy-saving overachievement	PAT, Energy Conservation Act, BEE	Not a general carbon allowance
REC	Renewable-electricity environmental attribute	CERC REC Regulations, 2022	Not automatically an offset
CCC	One tCO ₂ e reduction/removal within CCTS rules	BEE administration, GCI registry, CERC trading oversight	Market existence does not prove mature liquidity

5. CCTS dated status

- CCTS notified 28 June 2023; offset mechanism added 19 December 2023.
- Compliance arm uses product-linked GHG-emission-intensity targets.
- October 2025: 282 obligated entities in aluminium, cement, chlor-alkali and pulp/paper.
- 13 January 2026: 208 more in petroleum refineries, petrochemicals, textiles and secondary aluminium; total 490.
- 26 June 2026 iron-and-steel target notification remained draft in evidence checked at cutoff.
- Never cite an unofficial current certificate price or trading volume.

6. Climate-finance vocabulary

- Public, private and alternative sources can support mitigation and adaptation.
- Keep pledge, commitment, approval, provided finance, mobilised private finance, disbursement, expenditure and verified outcome separate.
- Concessionality = terms more favourable than market terms.
- Additionality = investment or climate result beyond the counterfactual.
- Blended finance uses targeted public/philanthropic risk capital to mobilise commercial capital.
- Adaptation has a bankability gap because avoided-loss benefits are often diffuse and non-cash-flow generating.

7. Green bonds, sovereign green bonds and deposits

- Green bond = debt with earmarked eligible environmental use of proceeds.
- SGrB framework approved 9 November 2022.
- FY 2023-24: Rs 20,000 crore issued; eligible transfers Rs 20,785.60 crore; Rs 785.60 crore above proceeds came from general revenue.
- Sector transfers: Rs 16,087.99 crore clean transport; Rs 4,578.47 crore renewable energy; Rs 119.14 crore living resources/land use.
- Greenium requires a matched conventional benchmark; it is not proof of impact.
- RBI's 28 November 2025 directions govern green deposits for covered commercial banks, SFBs and deposit-taking NBFCs/HFCs.
- Board policy, financing framework, allocation assurance and impact assessment matter.

8. Financial-sector climate risk

- Physical risk: acute/chronic hazard -> asset, income, collateral and operational loss.
- Transition risk: policy/technology/preference change -> repricing and stranded assets.
- Liability risk: claims and compensation from harm or misleading disclosure.
- Systemic channels: concentration, insurance gaps, correlated defaults and fire sales.
- RBI climate disclosure framework dated 28 February 2024 remained draft at cutoff.
- March 2025: RBI said disclosure finalisation and scenario/stress-testing guidance were under development.

9. SEBI and taxonomy

- BRSR is SEBI's listed-entity sustainability report; BRSR Core is a KPI subset.
- FY 2025-26 BRSR Core assessment/assurance glide path: top 500 listed entities.
- FY 2026-27: top 1,000.
- Value-chain ESG disclosure voluntary from FY 2025-26; assessment/assurance voluntary from FY 2026-27.
- Green-debt rules require objectives, selection, proceeds tracking, utilisation and impact disclosure.
- DEA climate-finance taxonomy was a draft dated 7 May 2025; no final framework identified by cutoff.
- Greenwashing chain: label -> eligibility -> allocation -> output -> outcome -> additionality.

10. Circular economy

- Hierarchy: refuse/redesign -> reduce -> reuse -> repair -> refurbish -> remanufacture -> recycle -> recover -> dispose.
- LCA requires functional unit, system boundary, inventory and impact method.
- Track both material intensity and absolute throughput because rebound can offset gains.
- Industrial symbiosis uses one firm's by-product, heat or water as another's input.
- EPR shifts specified end-of-life obligations upstream but can be met through authorised systems.
- Dated EPR streams: waste tyres 21 July 2022; batteries 22 August 2022; plastic-packaging guidelines February 2022; e-waste rules 2 November 2022 effective 1 April 2023; used oil amendment 18 September 2023.
- Integrate informal workers, occupational safety and traceable processing.

11. Trade and just-transition boundary

- EU CBAM definitive phase began 1 January 2026; it is not part of CCTS.
- Embedded-emissions MRV matters for exporters; recognition of Indian carbon costs cannot be assumed.
- Cross-link Topic 20 for WTO and trade-law analysis.
- Just transition covers workers, regions, consumers, participation, skills, income support and affordable energy.

12. Mains evaluation spine

1. Define the externality, finance instrument or material loop.
2. Identify authority, legal perimeter, date and implementation stage.
3. Trace price, quantity, balance-sheet or lifecycle mechanism.
4. Test MRV, additionality, leakage, greenwashing and rebound.
5. Evaluate absolute emissions, resilience, material throughput, jobs, affordability and fiscal risk.
6. Conclude with carbon price + standards + innovation + public investment + social protection.